Financial Management

day / date:

Chapter : 7 Bonds & their valuation

Value of Bond = $V_B = \frac{FV}{(1+r)^n} + P\left[\frac{1-(1+r)^{-n}}{r}\right]$

Yield to maturity = $YTM = \frac{C + \frac{FV-PV}{n}}{\frac{FV+PV}{2}}$ or Current yield + Capital gain yield.

Capital gain yield = $\frac{P_1 - P_0}{P_0}$ or $r_s - \frac{D_1}{P_0}$ → dividend yield

Current yield = $\frac{\text{Coupon payment}}{\text{selling price}}$

Yield to call = $\frac{C + \frac{FV-CV}{n}}{\frac{FV-CV}{2}}$

Console = $\frac{\text{Int. PMT}}{\text{Discount rate}}$

Chapter : 9 Stocks & their Valuation

→ Stock / Equity Valuation

Step ① FCFF / FCFE → D_0

FCFF = NI + dep exp − ΔNOWC − capital expanditure (ΔNOWC ← CA − CL)

FCFE = NI + dep exp − ΔNOWC − capital expanditure − debt repayment + debt issuance

Step ② Calculate r_s → 3 methods [if not then WACC]

* CAPM — $r_s = r_f + (r_m - r_f)\beta_L$ ∵ $D_1 = D_0(1+g)$
* Retained earning (DCF) — $r_s = \frac{D_1}{P_0} + g$
* Bond yield + risk premium

Step ③ → ① Discount model − $P_0 = \frac{D_1}{(1+r_s)^1} + \frac{D_2}{(1+r_s)^2} + \dots + \frac{D_n}{(1+r_s)^n}$

② Declining growth model − $P_0 = \frac{D_1}{(1+r_s)^1} + \dots + \frac{D_n}{(1+r_s)^n}$ — same

③ Constant growth model

④ − $P_0 = \frac{D_1}{r_s - g}$ $g = (1 - \text{Div}\ P_0) \times ROE$

↳ discount $\frac{P_0}{(1+r_s)^n}$ (n) → Previous year

Zero growth model

$P_0 = \frac{D_0}{r_s}$ ∵ $D_0 = \frac{D_1}{(1+g)}$

Last Step : Total Value (Sum of all)

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Dividend yield = $\frac{D_1}{P_0}$

Capital gain yield = $r_s - \frac{D_1}{P_0}$

Horizon / continuing value =

T.V = $D_0(1+g) = D_1$

H.V = $\frac{\cancel{D_1}}{\cancel{(1+r_s)}}$ $\frac{D_1}{r_s - g} = P_0$

Discounted = $\frac{P_0}{(1+r_s)^{n}}$ + previous year

Per share = $\frac{\text{Price}}{\text{No. of shares}}$

Chapter 10 Cost of Capital

WACC = $w_d r_d (1-T) + w_p r_p + w_c r_c$

Cost of preferred stock = r_p = $\frac{\text{Dividend price}}{\text{Preferred price}} = \frac{D_p}{P_p}$

Cost of common stock = r_s → CAPM

$r_s = r_f + (r_m - r_f)\beta_L$

r_s = Bond yield + risk premium

DCF or retained earning

expected / future dividend

$r_s = \frac{D_1}{P_0} + g$ ∵ $D_1 = D_0(1+g)$

current / last dividend

$$r_d = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

$$r_s = \frac{D_1}{P_0(1-F_c)} + g$$

Net Present Value = NPV = $\frac{CF_t}{(1+i)^t}$ $\boxed{i = WACC}$ [cashflow is different.]

Internal Rate of return = IRR = $\left[\frac{CF_1}{(1+x)^1} + \frac{CF_2}{(1+x)^2} + \frac{CF_3}{(1+x)^3}\right]$

{Shift + Solve (CALC) Solve for (0)

Modified IRR = MIRR = $\left(\frac{\text{Terminal value}}{CF_0}\right)^{1/n} - 1$

Payback Period = PBP = no. of years prior recovery + $\frac{\text{unrecovered cost}}{\text{CF at the year of fall recovery}}$

Discounted payback period = Same formula ↑

Cash flow estimations → New project
- Step 1 ΔNOWC
- Step 2 Initial cash oulay CF_0
- Step 3 Depreciation
- Step 4 Calculate cashflows
- Step 5 Terminal cashflow

↓

Replacement project
- Step 1 Initial cashflow CF_0
- Step 2 Revenue
- Step 3 Depreciation
- Step 4 Opportunity cost
- Step 5 CF estimation

Replacement chain →
- Step 1 Find NPV of each
- Step 2 equalize CF
- Step 3 NPV (extended)
- Step 4 EAA

Equivalent annual annuity = EAA = $\frac{r\,(NPV)}{1-(1+r)^{-n}}$

Scenario Analysis → ① Return / Expected NPV
$= P_1(NPV_1) + P_2(NPV_2) + \ldots$

② Risk = $\sigma = \sqrt{P_1(x_1-\bar{x})^2 + P_2(x_2-\bar{x})^2 + \ldots}$

③ Coefficient of variance = CF = $\frac{\sigma}{\text{Return}}$

Chapter: 14 ~~Real options . . . Budgeting~~ capital structure and leverage

Optimal Capital Budget

States	Prob	EBIT	(EBIT − Int. exp)(1 − T)	ROE	P(ROE)	$P(ROE - \hat{ROE})^2$
					↓ sum $\hat{ROE}$	↓ sum σ^2 = underroot = σ

∵ Int. exp = $r_d \times D$

$CV = \frac{\sigma}{\hat{ROE}}$

Chapter: 14 Capital structure & Leverage

day / date:

Operating leverage →

① Return on invested capital = ROIC = (~~EBIT Int exp~~)

$$ROIC = \frac{EBIT(1-T)}{\text{investor supplied capital}}$$

② Return on equity = ROE = $\frac{NI}{\text{equity}}$

Equity = capital (1 − Debt to equity ratio)

NI = (EBIT − Int. exp)(1 − T)

Breakeven Analysis →

① Breakeven in units / points = $\frac{FC}{SP - VC}$

② Gain / loss = SP − VC − FC

③ Breakeven in sales = no. of units × SP

Hamadah's equation →

① WACC (existing)

② Levered Beta = β_L → through CAPM

$$\beta_L = \frac{r_s - r_f}{(r_m - r_f)}$$

③ Hamadah's equation = $\beta_L = \beta_U\left[1 + (1-T)\frac{\text{Debt}}{\text{equity}}\right]$

④ r_s → CAPM

$$r_s = r_f + (r_m - r_f)\beta_L$$

Recapitalization →

① Earnings per share = EPS = $\frac{NI}{\text{no. of shares outstanding}}$ or $\frac{(EBIT - Int.exp)(1-T)}{\text{no. of shares outstanding}}$

② Issues repurchase = $\frac{\text{Debt}}{\text{no. of shares}}$

③ Time interest expense = TIE = $\frac{EBIT}{Int.exp}$

↓

$r_d \times D$

day / date:

Chapter : 15 Distributions to Shareholders

Dividend = NI − (Target equity ratio × Target capital)

Div P_o = $\frac{DIV}{NI}$ (Dividend payout ratio.) — Dividend residual model

or Div = NI × P_o

Stock split ⟶

New dividend = Last year dividend × g

= R/E = NI (1 − P_o)

Total equity = (New investment)(1 − Debt to equity ratio)

Dividend per share = DPS = $\frac{\text{Total dividend}}{\text{shares outstanding}}$

Dividend yield = $\frac{DPS}{P_o}$

Stock repurchase ⟶

EPS = $\frac{NI}{\text{no. of shares outstanding}}$

Dividend Payout ratio = $\frac{DPS}{EPS}$

ROE = $\frac{NI}{\text{equity}}$

g = Retention × ROE

↓

(1 − Div P_o)

retained earning (DCF) = r_s = $\frac{D_1}{P_o} + g$

Total dividend = DPS × no. of shares

Market capitalization = No. of shares × stock price

Stock dividend to market capital = $\frac{\text{Total dividend}}{\text{Market capitalization}}$

New no. of shares = $\frac{T.D}{P_o}$

Dilution of EPS = Old EPS − New EPS

day / date:

Chapter : 16 Working capital management

$NWC = CA - CL$

$NOWC = (CA - CL) - \text{Note payables}$

Duopond Analysis of ROE $= \dfrac{NI}{\text{equity}}$

Cash Conversion Cycle = Inventory conversion + receivables collection - Payable defferal

$$CCC\ (\text{in days}) = \frac{\text{Inventory}}{COGS/365} + \frac{\text{receivables}}{\text{sales}/365} - \frac{\text{Payables}}{COGS/365}$$

$$\text{Inventory turnover} = \frac{COGS}{\text{inventory}}$$

$$\text{Nominal Cost of Trade Credit} = \frac{DR}{100\% - DR} \times \frac{365}{\text{no. of trade period (Total)} - DP}$$

Effective Cost of Trade credit

(1) Company

$$= \left[1 + \frac{DR}{100\% - DR}\right]^{\frac{365}{n-DR}} - 1$$

(2) Bank $= \left[1 + \dfrac{i_{nom}}{m}\right]^{m} - 1$

Cash Budget ⟶ (1) Collections

(2) Payments

(3) Net cash gain or loss